

The Impact of Outliers: The Raisin River Canoe Race | SOLUTIONS

Downriver racing is a sport where competitors take their watercraft (usually a canoe, kayak, or stand-up paddle board) down a race course in competition with other paddlers, sometimes navigating rapids, in an attempt to get the fastest time either overall or in their class (classes are split up by boat and paddler type). The Raisin River Canoe race has been running for over 50 years on the Raisin River in Eastern Ontario. The usual race course is 30 km (about 18.6 miles) and contains rapids that fluctuate in difficulty with the water level. In race results, a DNF (did not finish) refers to a boat that did not make it to the finish line. Most of the time, these boats are paddlers who at some point decided to drop out of the race. This is most common in longer races (like Raisin River) or races with dangerous rapids. While the rapids on the Raisin River course are not too intimidating, as water level increases, the difficulty (or, more importantly, the fear factor) increases as well. Race directors are responsible for keeping track of their competitors and making sure they get off the river safely, so it is important that race officials are aware of possible DNFs. Having a way to predict the ratio of participants that may DNF the race based on the water level (information that is known ahead of time) could be very useful. For this activity we will be using data containing DNF and water level measures from the Raisin River Race from 2005-2025 to look at how outliers can impact a model.³

```
library(tidyverse)
library(dplyr)
library(broom)

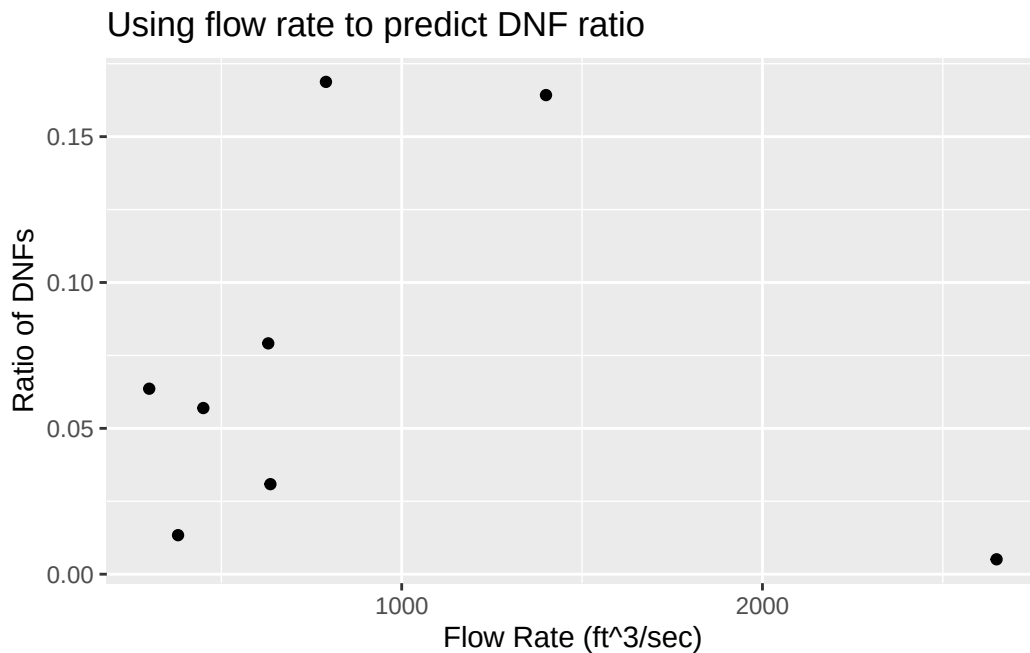
RR_dnf_data <- read_csv("raisin_river_DNF.csv")
```

This data set contains flow measurements (indicative of water level) in ft^3/sec and the ratio of DNFs (# of DNFs/total participants) from the Raisin River Race from 2005-2025 (excluding, 2020 and 2021, when the race didn't run, and 2019, which has incomplete data).

Explore the relationship between flow rate and ratio of DNFs

1. Graph a scatterplot that shows flow rate as a predictor for rate of DNF

```
ggplot(RR_dnf_data, aes(x = flow, y = ratio_DNF)) +  
  geom_point() +  
  labs(x = "Flow Rate (ft3/sec)",  
       y = "Ratio of DNFs",  
       title = "Using flow rate to predict DNF ratio")
```



2. Fit a model using flow rate as a predictor for ratio of DNFs, print a summary of the model, and meaningfully interpret the slope coefficient.

```
dnf_mod <- lm(ratio_DNF ~ flow, RR_dnf_data)  
summary(dnf_mod)
```

Call:

```
lm(formula = ratio_DNF ~ flow, data = RR_dnf_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.06308	-0.04668	-0.01622	0.02708	0.09521

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.914e-02	3.809e-02	2.077	0.083 .
flow	-7.056e-06	3.273e-05	-0.216	0.836

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.06783 on 6 degrees of freedom

Multiple R-squared: 0.007689, Adjusted R-squared: -0.1577

F-statistic: 0.04649 on 1 and 6 DF, p-value: 0.8364

a) The model has a negative coefficient for flow's impact on DNF ratio. Does this make sense with what you saw on the graph from question 1?

It does not make sense. On the scatterplot, there is a positive trend.

b) Note the p-value for the flow coefficient. Is it significant?

No, the p-value for the flow coefficient is 0.836, which means there is no evidence for flow being a useful predictor in the model made using this data set.

c) Comment on the appropriateness of the model

Based on the scatterplot and the model summary, the model is not appropriate. There is one observation that is influencing the model too much for it to make sense.

Investigate Influential Observations

3. Use the augment function to add the model's residuals into the dataset

```
dnf_resid = left_join(select(RR_dnf_data, year, ratio_DNF, flow), augment(dnf_mod))
```

Joining with `by = join\_by(ratio\_DNF, flow)`

a) Do any observations have unusual leverage? standardized residual? influence?

```
dnf_resid |> filter(.hat > 2*(2/8))
```

```
# A tibble: 1 x 9
  year ratio_DNF flow .fitted .resid .hat .sigma .cooks .std.resid
<dbl>   <dbl> <dbl>   <dbl>   <dbl> <dbl> <dbl>   <dbl>   <dbl>
1  2017   0.00510 2649   0.0604 -0.0553 0.833 0.0429   10.0   -2.00
```

2017 has leverage greater than $2/8$ ($8 = n$), making it unusual.

```
dnf_resid |> filter( abs(.std.resid) >= 2)
```

```
# A tibble: 0 x 9
# i 9 variables: year <dbl>, ratio_DNF <dbl>, flow <dbl>, .fitted <dbl>,
#   .resid <dbl>, .hat <dbl>, .sigma <dbl>, .cooks <dbl>, .std.resid <dbl>
```

All observations have standardized residuals with absolute value less than 2. There are no observations with unusual standardized residuals.

```
dnf_resid |> filter(.cooks > 1) ## 2017 is highly influential
```

```
# A tibble: 1 x 9
  year ratio_DNF flow .fitted .resid .hat .sigma .cooks .std.resid
<dbl>   <dbl> <dbl>   <dbl>   <dbl> <dbl> <dbl>   <dbl>   <dbl>
1  2017   0.00510 2649   0.0604 -0.0553 0.833 0.0429   10.0   -2.00
```

2017 has a cook's distance of 10 (much greater than 1), making it highly influential.

Outlier Removal

4. We only want to remove outliers if we have reason to believe that outside factors may be influencing the data in a way that prevents data analysis. In this case, the outlying observation is from 2017. In 2017, the water level was dangerously high, causing the race officials to move the race start to Delaney Road (roughly 4 miles further down the river), leaving the finish in the normal place. The map below shows the race course. Do you think this change warrants removing the year from the data set? Why or why not?



Figure 1: Raisin River Race Course

Answers may vary, but the shortening of the race course reduces the difficulty of the race notably, even with high water. This decrease in difficulty (or more importantly, effort, especially with the very fast-moving water) can account for a drop in the DNF ratio. Because of this impact, removing this observation from the data set makes sense, as all other years represent the original race course, not the shortened version.

5. Remove the influential observation from the data set.

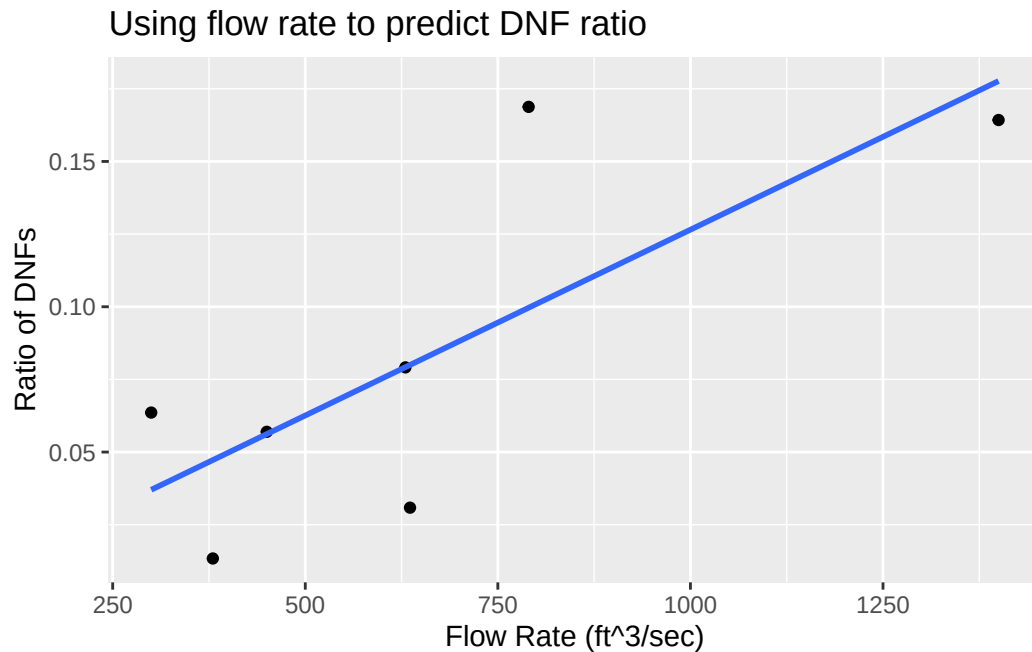
```
RR_dnf_data2 <- RR_dnf_data |> filter(year != 2017)
```

6. Regraph the scatterplot without the influential observation and add a smoother.

```
ggplot(RR_dnf_data2, aes(x = flow, y = ratio_DNF)) +  
  geom_point() +  
  geom_smooth(method = lm, se = FALSE) +  
  labs(x = "Flow Rate (ft3/sec)",
```

```
y = "Ratio of DNFs",
title = "Using flow rate to predict DNF ratio")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



7. Refit the model with the new version of the data set and print a summary of the model.

```
dnf_mod <- lm(ratio_DNF ~ flow, RR_dnf_data)
summary(dnf_mod)
```

Call:

```
lm(formula = ratio_DNF ~ flow, data = RR_dnf_data)
```

Residuals:

Min	1Q	Median	3Q	Max
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Residual standard error: 0.06783 on 6 degrees of freedom
Multiple R-squared: 0.007689, Adjusted R-squared: -0.1577
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a) Meaningfully interpret the slope coefficient.

For every 100 ft³/sec increase in flow, we can expect a 0.01279 unit increase in the ratio of DNFs.

b) If the flow level is 1100 ft³/sec and there are 200 competitors about how many DNFs can we expect?

```
-0.001343 + (0.0001279*1100)
```

```
[1] 0.139347
```

```
200 * 0.139347
```

```
[1] 27.8694
```

If flow level is 1100 ft³/sec and there are 200 competitors, we can expect about 27 DNFs.